

Acoustic Guitar Stool

Measure for Measure — Anthropometric Furniture & Instrument Holster

A bespoke timber stool and integrated guitar holster designed around my own anthropometric dimensions. Tasked with creating a 1:1 furniture piece tailored to personal ergonomics, I focused on eliminating posture fatigue during long acoustic guitar playing sessions.

Discipline

Furniture & Joinery

Context

CSM 'Measure for Measure'

Tools

Fusion 360, Woodshop

Outcome

A full-scale structural pine stool with an integrated side holster and lower amplifier shelf, translating 1:5 stripwood models and orthographic CAD drawings into physical joinery.

Client profile & ergonomic mindmap



Fig. 01 — Initial brainstorm mindmap exploring personal daily seated rituals, posture pain points, and equipment storage

Designing for Myself as the Primary Client

The 'Measure for Measure' brief asked us to design and build a 1:1 structural furniture piece calibrated to our own body measurements and daily habits.

Mapping my routines in a shared student room highlighted three activities where standard chairs fail: acoustic guitar practice, extended reading, and desk gaming.

01 **Postural Slouching** — Low beds and soft chairs force a rounded lower back when leaning over an acoustic guitar.

02 **Obstructed Right Arm** — Standard chair armrests collide with the lower bout of a dreadnought guitar body.

03 **Instrument Clutter** — Leaving a guitar leaning against a wall or amplifier between practice sessions risks knocks.

Task 1 — personal needs mapping

01 — Guitar Playing (Chosen)
Requires an open right-hand side for strumming clearance, a shallow perched seat to keep the spine upright, and an integrated neck holster + lower amp shelf.

02 — Reading
Long reading sessions cause numb limbs unless seat angle and lumbar support work together.

03 — Gaming
Most gaming chairs rely on aggressive racing-seat clichés rather than warm domestic timber.

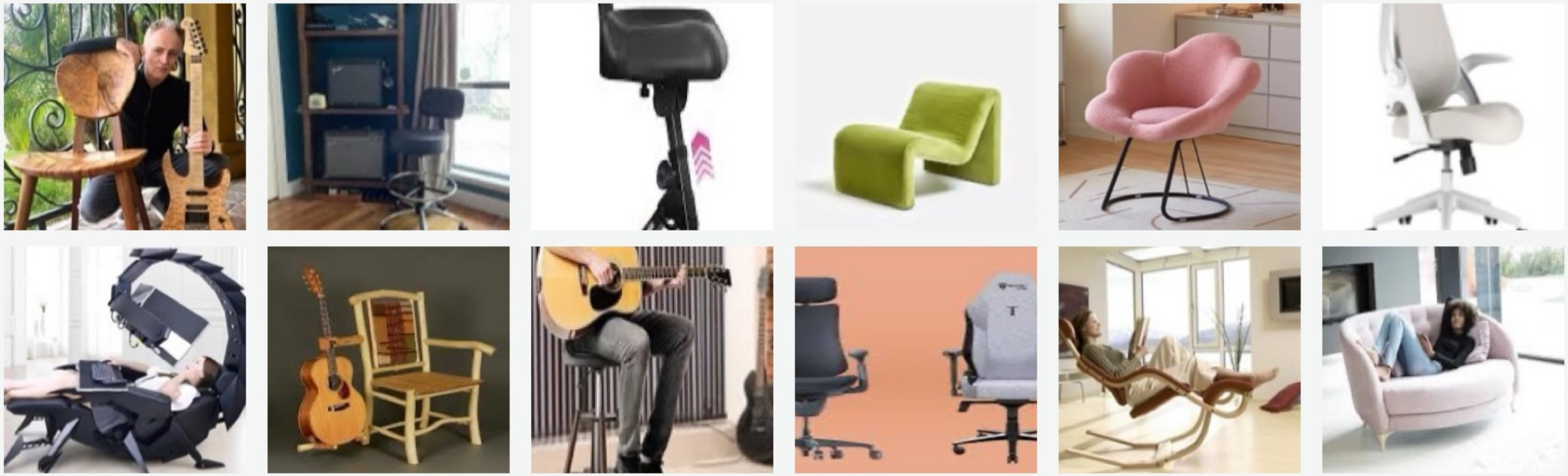


Fig. 02–13 — Primary observation photos and posture sketches across guitar playing, reading, and gaming

Task 2 — critical dimensions & anthropometrics





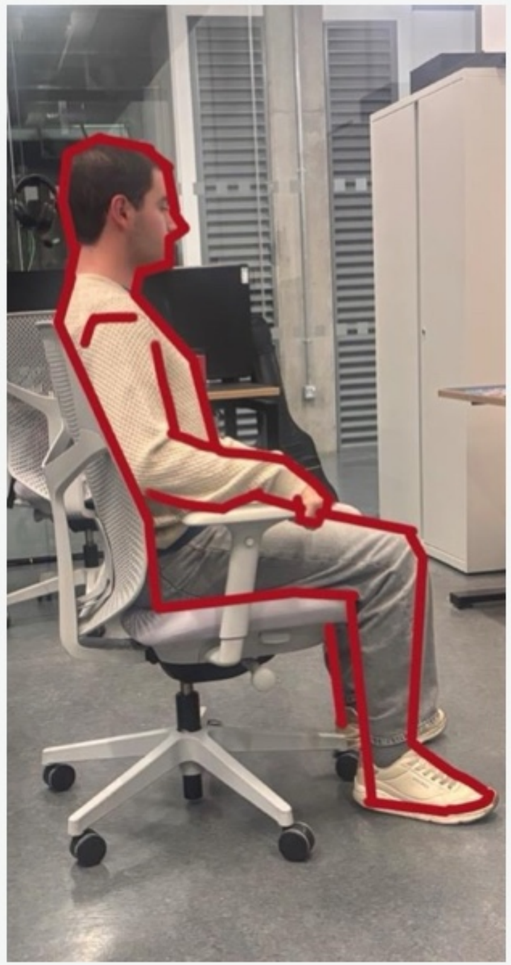


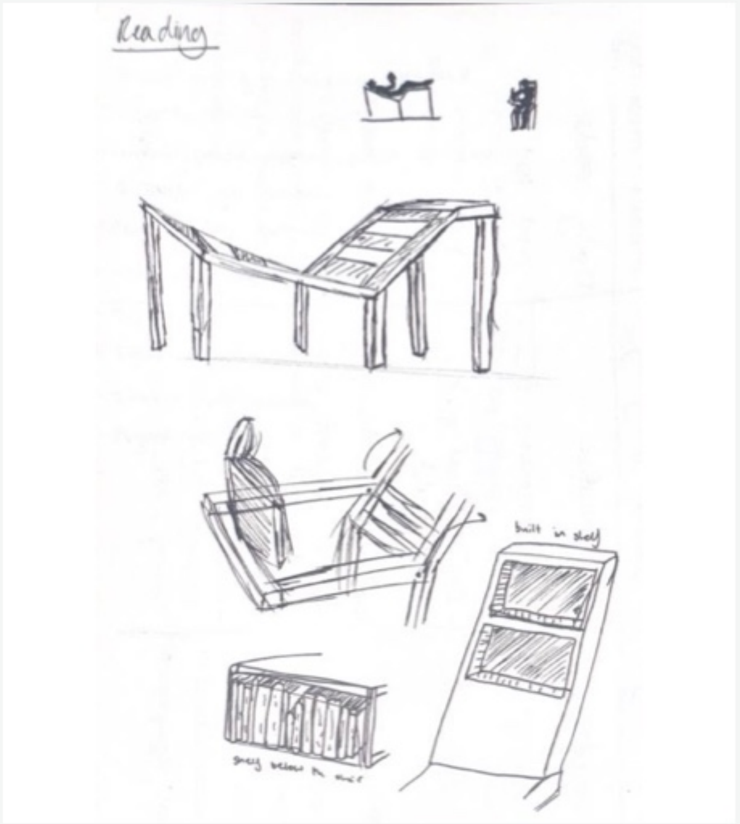
Fig. 14–16 — Seated anthropometric posture analysis with red joint-angle overlays

Calibrating Ergonomics to My Body & Instrument

By photographing my seated playing posture against a metric reference grid, I measured the exact seat height (420mm) and shallow perched depth (200mm) needed to maintain an open hip angle while keeping the guitar waist stable on the right thigh.

#	Critical Dimension	My Target	Standard Chair	Ergonomic Rationale
1	**Seat height**	420 mm	~430 mm	Keeps thighs level with feet flat on floor or foot rail
2	**Seat depth**	200 mm	~400 mm	Shallow perch prevents slouching into a deep backrest
3	**Seat width**	400 mm	~440 mm	Compact footprint with open right side for guitar body
4	**Left side/back rest**	130–600 mm	~480 mm	Supports fretting-hand shoulder without blocking right elbow
5	**Holster height**	650 mm	~220 mm	Cradles acoustic guitar neck upright beside the seat

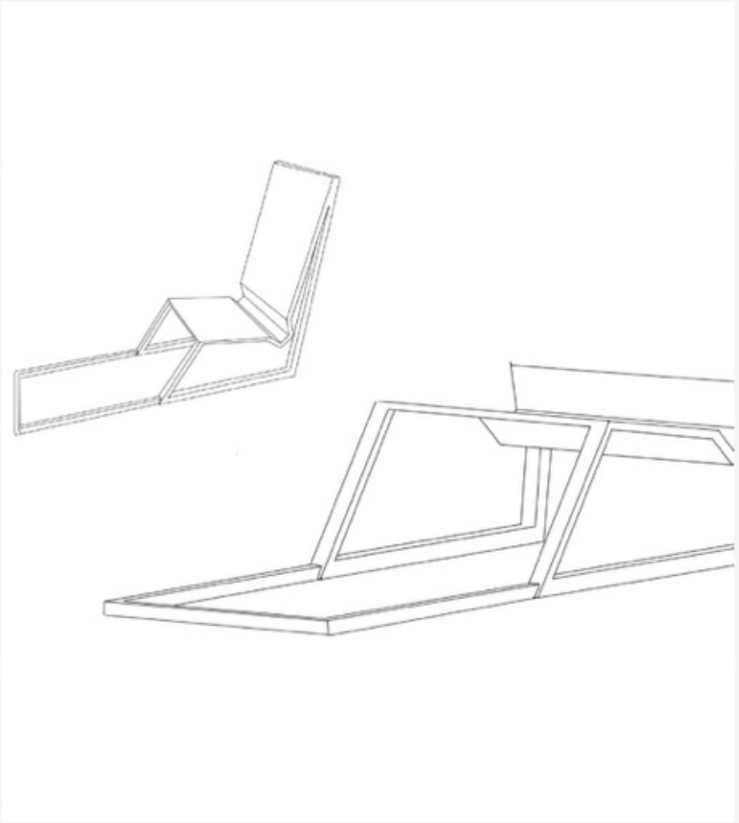
Task 3 — initial concept sketches



Concept Sheet 01 — Asymmetric Backrests
Explored wrap-around left-hand rests that leave the right strumming arm completely unobstructed.

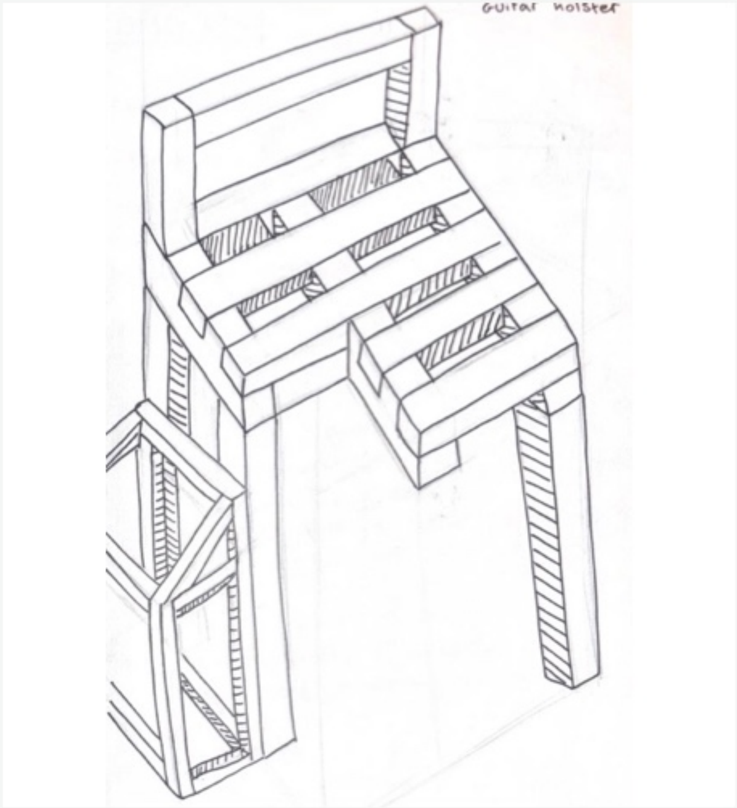


Concept Sheet 02 — Slatted & Cantilevered Frames
Investigated slatted timber seats with under-seat amplifier storage and side-mounted neck cradles.

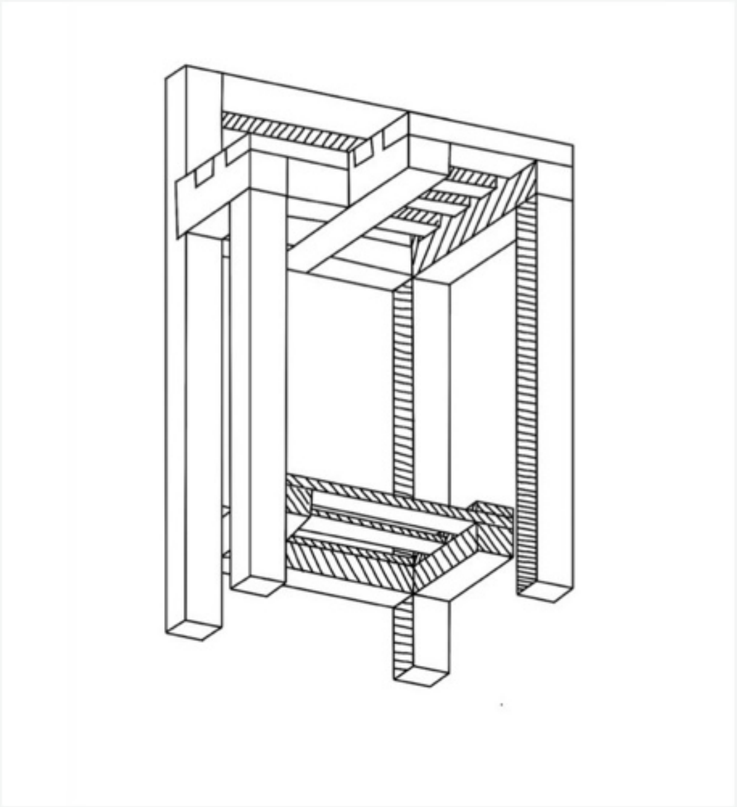


Concept Sheet 03 — Angular & Folding Geometries
Tested triangular A-frame legs and notched cross-rails for torsional stiffness under playing movement.

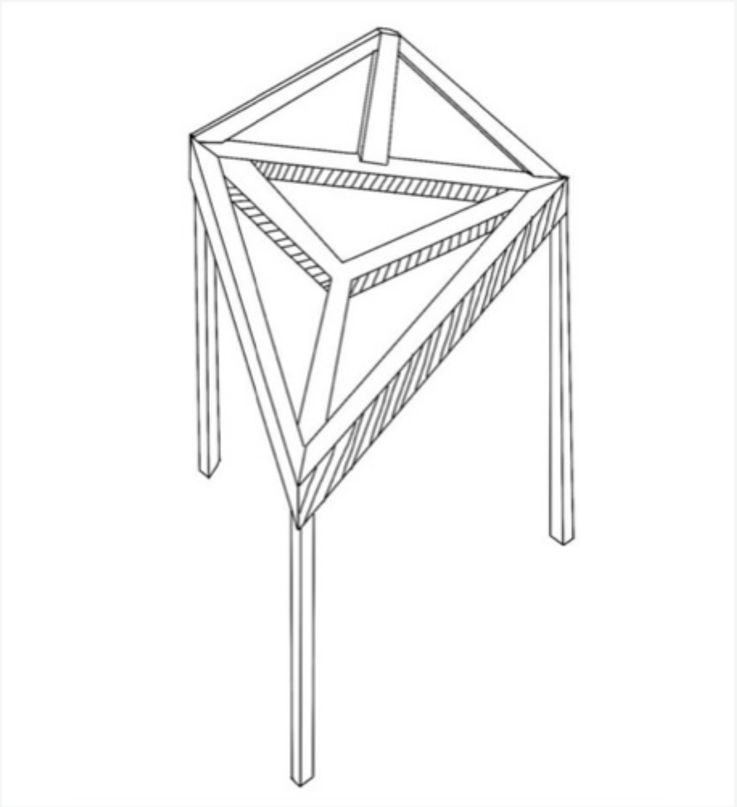
Task 3 — isometric joinery & structural sketches



01 — Side-Rest Frame Study
Orthogonal frame with elevated left armrest and lower cross-stretcher for foot elevation.



02 — Slatted Stool + Holster (Chosen)
Interlocking 44×44mm legs and 44×22mm slats with a cantilevered right-hand guitar neck yoke and amp shelf.



03 — Acute Triangular Frame
Explored raked triangular legs meeting at acute mitre joints beneath a narrow saddle seat.

Task 3 — 1:5 scale stripwood models



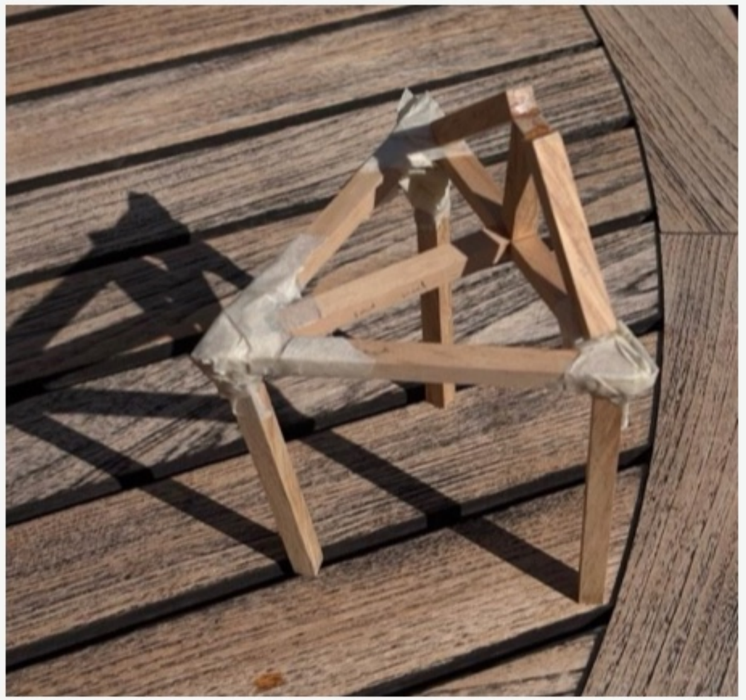
Model 01 — Side-Rest Frame (1:5)

Clean rectangular frame supporting the left side while playing, but lacked dedicated guitar or amp storage.



Model 02 — Slatted Stool + Holster (Chosen)

Proved the most stable and functional: combines a rigid box-frame footprint, lower amp shelf, and cantilevered guitar neck holster.



Model 03 — Triangular Frame (1:5)

Building at 1:5 scale immediately revealed that the acute mitre joints had minimal glue surface area and twisted under load.

Making plan & timber cut list

Stock Allocation & Saw-Kerf Planning

Every structural component was nested across six 1200mm pine lengths—three heavy 44 × 44 mm square sections for primary legs and load-bearing rails, and three 44 × 22 mm rectangular sections for seat slats, shelf rails, and the guitar holster.

Board	Stock Section	Allocated Components & Cut Lengths	Used	Offcut
Board 1	**44 × 44 × 1200 mm**	2 × Rear upright legs @ 585 mm each	1,170 mm	~30 mm
Board 2	**44 × 44 × 1200 mm**	2 × Front legs @ 406 mm + 1 × Left side rail @ 356 mm	1,168 mm	~32 mm
Board 3	**44 × 44 × 1200 mm**	2 × Primary seat bearer rails @ 400 mm + 1 × Right holster arm @ 200 mm	1,000 mm	~200 mm
Board 4	**44 × 22 × 1200 mm**	2 × Seat/backrest slats @ 400 mm + 1 × Left stretcher @ 378 mm	1,178 mm	~22 mm
Board 5	**44 × 22 × 1200 mm**	2 × Seat slats @ 400 mm + 1 × Lower amp shelf rail @ 312 mm	1,112 mm	~88 mm
Board 6	**44 × 22 × 1200 mm**	1 × Front seat slat @ 400 mm + 1 × Right rail @ 244 mm + 1 × Holster jaw @ 220 mm	864 mm	~336 mm

Total stock utilisation: 3,338 mm / 3,600 mm (92.7%) for 44×44mm pine · 3,154 mm / 3,600 mm (87.6%) for 44×22mm pine

Joinery schedule & joint hierarchy

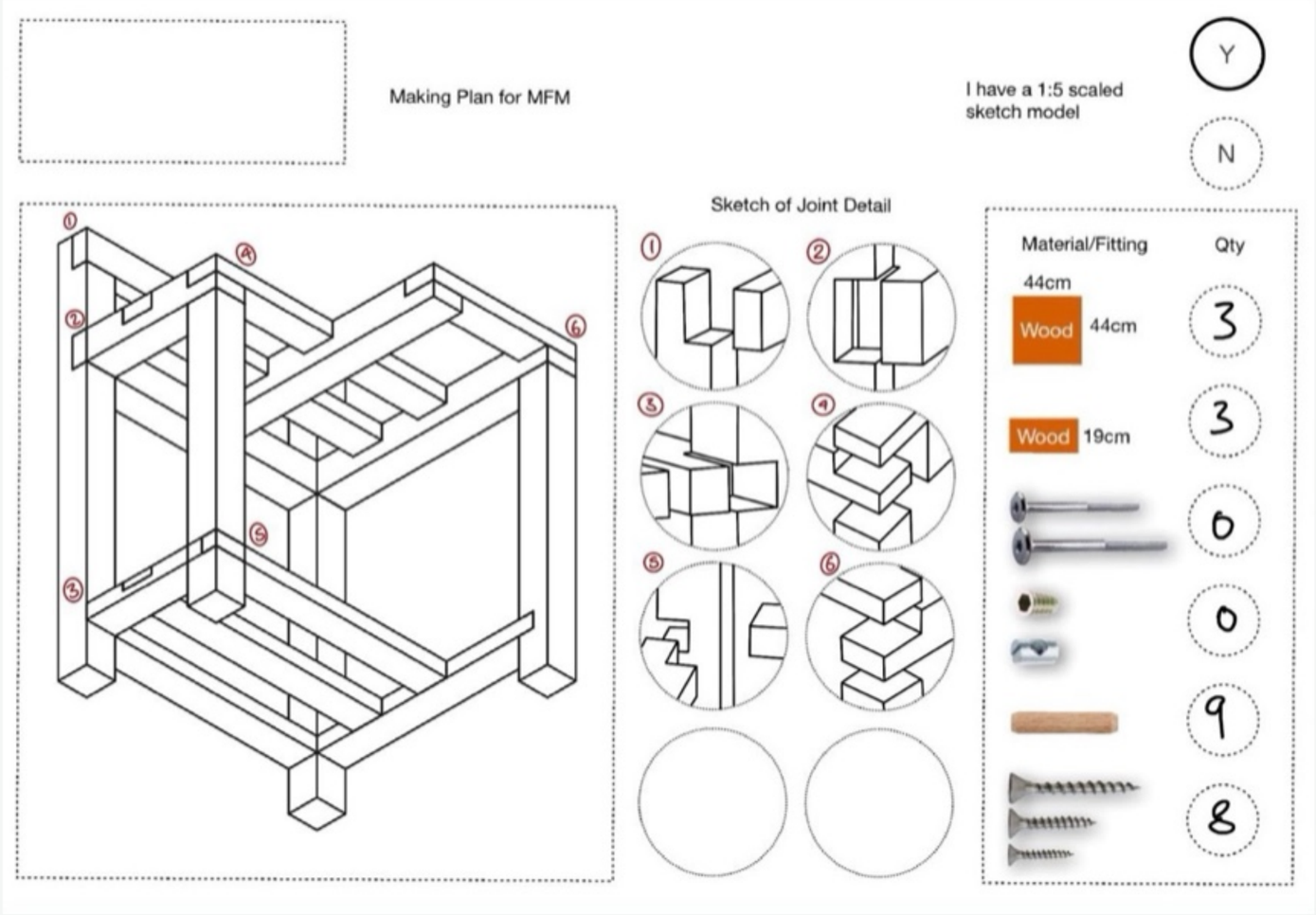


Fig. 23 — Joinery Sheet: numbered isometric joint key (1–6), cross-halving / lap-joint details, and dowel/screw schedule

Structural Joint Hierarchy

To resist racking when shifting weight with a guitar, the frame combines recessed housing/lap joints with mechanical fasteners.

01 **Joints 1 & 2 — Leg-to-Rail Lap Joints** — 22mm deep cheek trenches cut into the 44×44mm legs create positive mechanical shoulders supporting vertical seat loads.

02 **Joints 3 & 4 — Cross-Halving Seat Slats** — Notched 44×22mm seat slats lock across the primary bearers to prevent lateral twisting.

03 **Joints 5 & 6 — Cantilevered Guitar Yoke** — Reinforced with 8mm fluted hardwood dowels and countersunk steel wood screws to hold the cantilevered neck cradle.

CT pack part A — dimensioned orthographic drawings

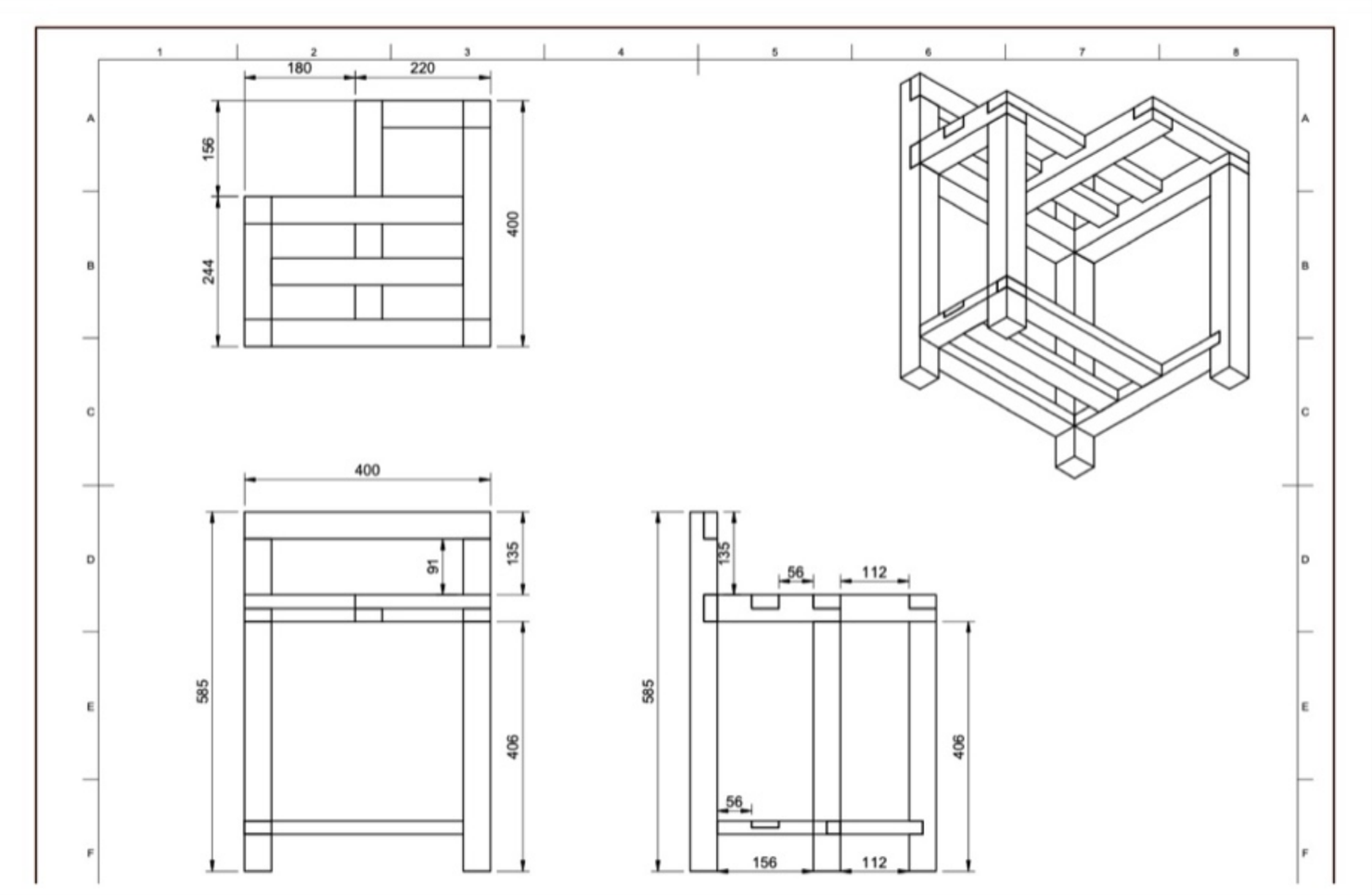


Fig. 24 — Third-angle orthographic projection sheet (Plan, Front, Side & Isometric) with overall dimensions in mm

Orthographic Verification

Drafting the complete stool in Fusion 360 confirmed critical clearances before cutting timber:

- 400 × 400 mm main frame footprint
- 585 mm overall rear height / 420 mm seat height
- 180 mm right-hand holster projection with a 91 mm padded neck slot
- 132 mm ground clearance beneath the lower amplifier shelf

CT pack part B — assembled & exploded CAD

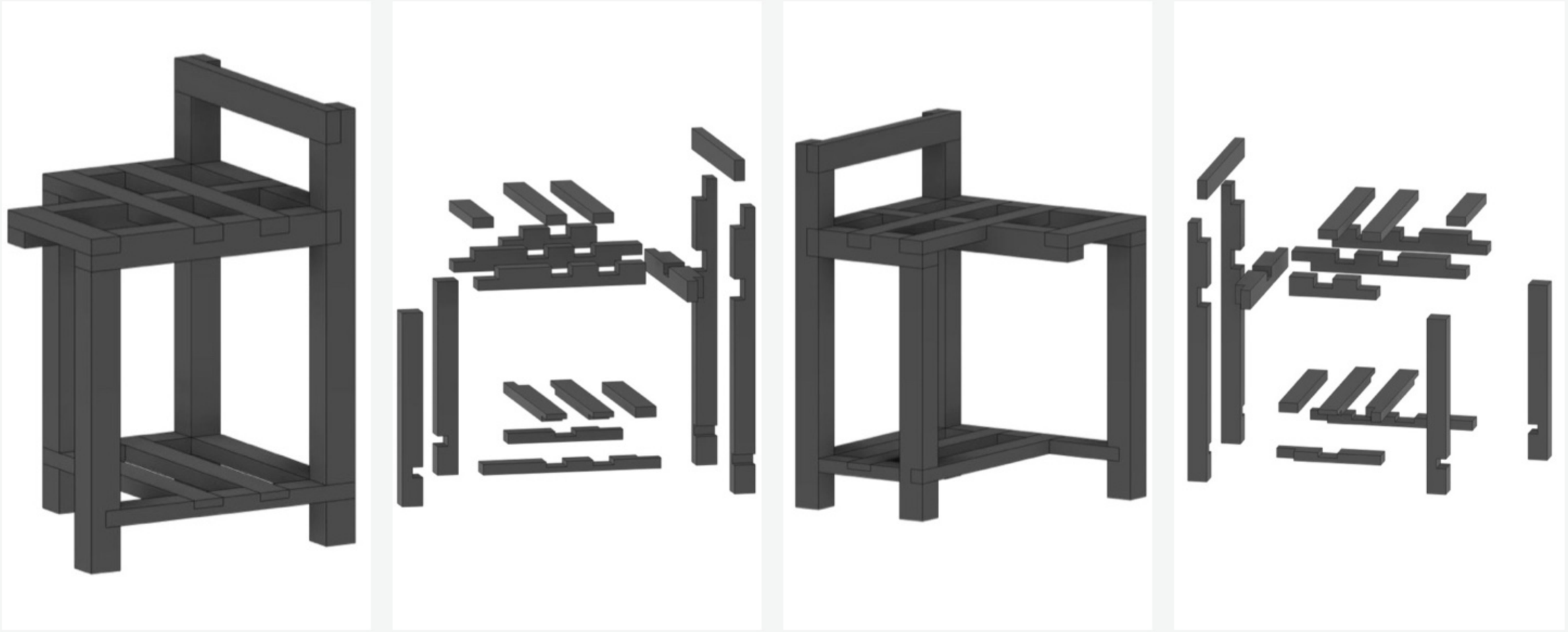


Fig. 25–28 — Fusion 360 assembled and exploded isometric views illustrating the 14 interlocking pine members

Making process part 1 — cutting & kerf calibration







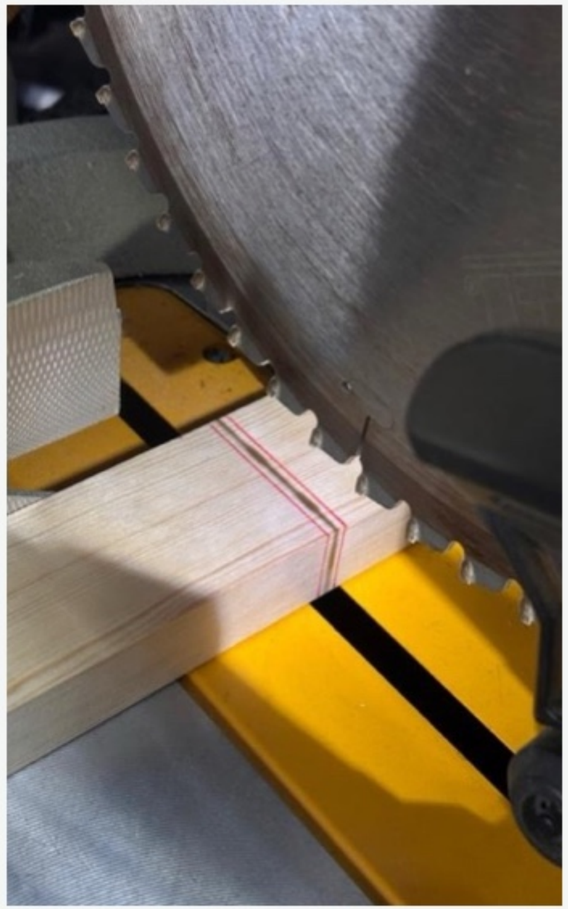


Fig. 29–32 — Referencing the 1:5 stripwood model at the chop saw, batch-cutting pine stock, verifying stepped lengths, and dry-laying joints

Making process part 2 — marking out & joint trenching

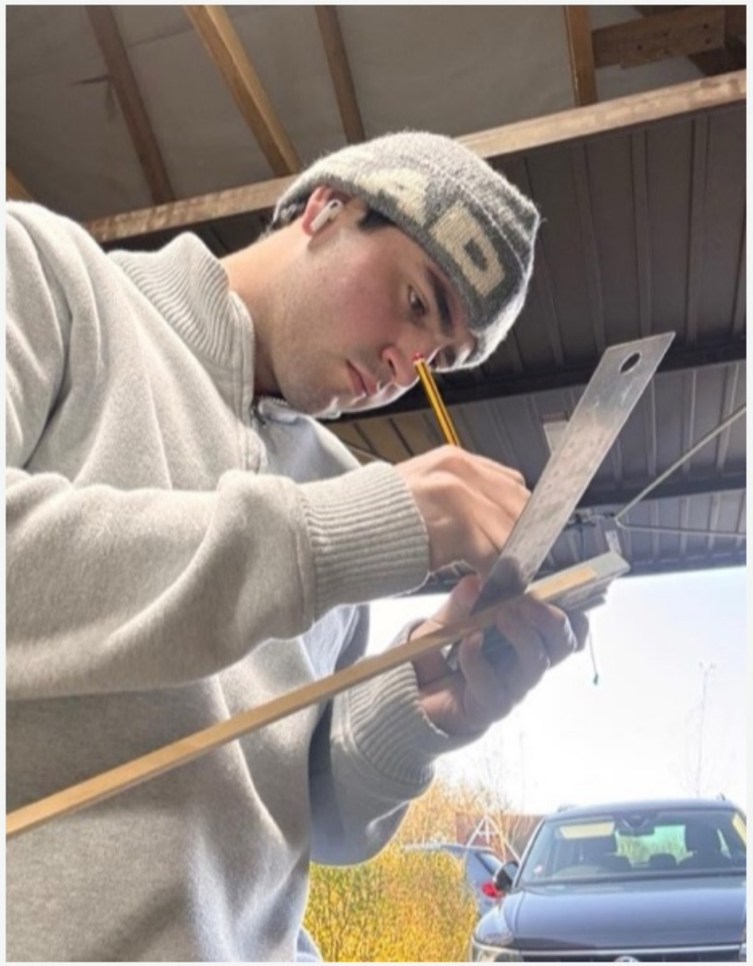




Fig. 33–34 — Transferring CAD dimensions onto 44×44mm pine with a try-square and batch-cutting housing trenches

Precision Marking & Shoulder Cutting

Each half-lap and housing joint was knifed and squared across all four faces of the 44mm pine stock.

To ensure consistent seat levelness across all four legs, mating rails were clamped side-by-side so shoulder lines could be struck simultaneously across the batch.

Workshop problem solving & project reflection

Challenges Encountered During Fabrication

Translating an intricate CAD model with dozens of interlocking notches into 44mm softwood within a tight workshop schedule surfaced three practical engineering lessons:

01 **Over-Complex Interlocking Notches** — My initial CAD featured housing trenches on every slat intersection. In the woodshop, hand-chiselling dozens of blind notches risked splitting the pine grain, so I simplified secondary slats to flush dowel/screw butt joints while retaining full lap joints on primary load bearers.

02 **Saw-Kerf Accumulation** — Even 2–3mm of accumulated kerf drift across a 1200mm board altered the final rail lengths; I adapted by re-referencing each cross-rail directly against the assembled leg frames.

03 **Holster Cantilever Rigidity** — Reinforced the projecting guitar neck yoke with through-dowels and structural screws so it cannot twist when a heavy acoustic guitar is docked.

Reflection & Future Iteration

The finished stool validated my anthropometric targets: the 420mm height and 200mm perched seat depth immediately encourage an upright spine while leaving the right strumming arm completely free.

In a next iteration, I would line the inside of the timber guitar holster with 4mm natural cork or vegetable-tanned leather to cushion nitro-cellulose guitar neck finishes, and machine the lap joints on a router table with a box-joint jig for sub-millimetre repeatability.

Final chair — studio photography







Acoustic Guitar Stool & Holster — Pine (44×44mm & 44×22mm), hardwood dowels, and steel fixings.

Ergonomic 420mm seat height · 200mm perched depth · Integrated side guitar neck holster & lower amp shelf.